

1. Calculation of carrier concentration of intrinsic semiconductor:-

An intrinsic semiconductor contains equal number of charge carriers i.e. electron and holes. Hence, the calculation of carrier concentration of intrinsic semiconductor involves measurement of electrons or holes numbers per unit volume of specimen in conduction and valance bands respectively.

a. Electrons concentration in intrinsic semiconductor:

In intrinsic semiconductor a number of free electrons are present in conduction energy band at different energy state with energy greater than the lowest conduction band energy E_c . The number of energy orbital per unit energy range is define as density of state. In three dimension, the density of state for electron above E_c is given by,

$$D_c(E) = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e}{\hbar^2}\right)^3 (E - E_c)} \quad \dots\dots\dots 1$$

Similarly, the probability for an electron to occupy in particular energy state E above fermi energy E_f is given by Fermi-Dirac distribution as

$$f_c(E) = \frac{1}{e^{\frac{E - E_f}{K_B T}} + 1} \quad \dots\dots\dots 2$$

For non degenerate case and at energy state $(E - E_f) \gg K_B T$ for electron, the distribution is modified as

$$f_c(E) = e^{-\frac{E - E_f}{K_B T}} \quad \text{by J.G. Siro}$$

Now, the concentration of electron in conduction band is

$$n = \int_{E_c}^{\infty} D_c(E) f_c(E) dE$$

from 1 and 2

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e}{\hbar^2}\right)^3} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{-\frac{E - E_c + E_c - E_f}{K_B T}} dE$$

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e}{\hbar^2}\right)^3} e^{-\frac{E_c - E_f}{K_B T}} \int_{E_c}^{\infty} (E - E_c)^{\frac{1}{2}} e^{-\frac{E - E_c}{K_B T}} dE$$

Put $\frac{E - E_c}{K_B T} = x$ or, $dx = \frac{dE}{K_B T}$ or, $dE = dx K_B T$

when $E = E_c$; $x = 0$
 $E = \infty$; $x = \infty$ then,

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e}{\hbar^2}\right)^3} e^{-\frac{E_c - E_f}{K_B T}} \int_0^{\infty} x^{\frac{1}{2}} (K_B T)^{\frac{1}{2}} e^{-x} (K_B T) dx$$

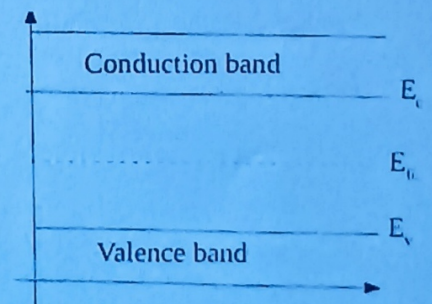


Fig: Fermi energy level lies between valence and conduction band in intrinsic semiconductor.

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e}{\hbar^2}\right)^3} (K_B T)^{3/2} e^{-\frac{E_c - E_f}{K_B T}} \int_0^\infty x^{\frac{1}{2}} e^{-x} dx$$

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e K_B T}{\hbar^2}\right)^3} e^{-\frac{E_c - E_f}{K_B T}} \int_0^\infty x^{\frac{1}{2}} e^{-x} dx$$

[being $\int_0^\infty x^n e^{-x} dx = n\Gamma n; \frac{1}{2}\Gamma \frac{1}{2} = \frac{1}{2}\sqrt{\pi}$]

$$n = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_e K_B T}{\hbar^2}\right)^3} e^{-\frac{E_c - E_f}{K_B T}} \frac{1}{2} \sqrt{\pi}$$

$$n = 2 \sqrt{\left(\frac{m_e K_B T}{2\pi\hbar^2}\right)^3} e^{-\frac{E_c - E_f}{K_B T}}$$

by S.G. Sir

$$n = N_c e^{-\frac{E_c - E_f}{K_B T}} \dots\dots\dots 3 \quad \text{where; } N_c = 2 \sqrt{\left(\frac{m_e K_B T}{2\pi\hbar^2}\right)^3}$$

This is required expression for the electron concentration in the conduction band of non-degenerate intrinsic semiconductor.

b. Holes concentration in intrinsic semiconductor:

The transition of electron from valence band to conduction band results to deficiency of electron in valence band. This deficiency of electron in valence band is called hole and it acts as positive charge carriers. In intrinsic semiconductor a number of holes are present in valence energy band at different energy state with energy lesser than the highest valence band energy E_v .

The number of energy orbital per unit energy range is define as density of state. In three dimension, the density of state for hole below E_v is given by,

$$D_h(E) = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} (E_v - E) \dots\dots\dots 4$$

Similarly, the probability for a hole to occupy in particular energy state E below fermi energy E_f is given by Fermi-Dirac distribution as

$$f_h(E) = 1 - \frac{1}{e^{\frac{E - E_f}{K_B T}} + 1}$$

$$f_h(E) = \frac{e^{-\frac{E - E_f}{K_B T}}}{e^{\frac{E - E_f}{K_B T}} + 1} \dots\dots\dots 5$$

For non degenerate case and at energy state $(E - E_f) \ll K_B T$ for hole, the distribution is modified as

$$f_h(E) = e^{-\frac{E - E_f}{K_B T}}$$

Now, the concentration of hole in valence band is

$$p = \int_{-\infty}^{E_v} D_h(E) f_h(E) dE$$

from 4 and 5

$$p = \int_{-\infty}^{E_v} \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3 (E_v - E)} e^{\frac{E - E_f}{K_B T}} dE$$

$$p = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} \int_{-\infty}^{E_v} (E_v - E)^{\frac{1}{2}} e^{\frac{E - E_f}{K_B T}} dE$$

$$p = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} \int_{-\infty}^{E_v} (E_v - E)^{\frac{1}{2}} e^{\frac{E - E_v + E_v - E_f}{K_B T}} dE$$

$$p = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} e^{-\frac{E_f - E_v}{K_B T}} \int_{-\infty}^{E_v} (E_v - E)^{\frac{1}{2}} e^{-\frac{E_v - E}{K_B T}} dE$$

Put $\frac{E_v - E}{K_B T} = x$ Or, $dx = \frac{-dE}{K_B T}$ Or, $dE = -dx K_B T$

when $E = E_v$; $x = 0$
 $E = -\infty$; $x = \infty$ then,

by s.c.sir

$$p = -\frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} e^{-\frac{E_f - E_v}{K_B T}} \int_{\infty}^0 x^{\frac{1}{2}} (K_B T)^{\frac{1}{2}} e^{-x} (K_B T) dx$$

$$p = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h}{\hbar^2}\right)^3} (K_B T)^{3/2} e^{-\frac{E_f - E_v}{K_B T}} \int_0^{\infty} x^{\frac{1}{2}} e^{-x} dx$$

$$p = \frac{1}{2\pi^2} \sqrt{\left(\frac{2m_h K_B T}{\hbar^2}\right)^3} e^{-\frac{E_f - E_v}{K_B T}} \int_0^{\infty} x^{\frac{1}{2}} e^{-x} dx$$

[being $\int_0^{\infty} x^n e^{-x} dx = n \Gamma n$; $\frac{1}{2} \Gamma \frac{1}{2} = \frac{1}{2} \sqrt{\pi}$]

$$p = N_v e^{-\frac{E_f - E_v}{K_B T}} \dots\dots\dots 6 \quad \text{where; } N_v = 2 \sqrt{\left(\frac{m_h K_B T}{2\pi\hbar^2}\right)^3}$$

This is required expression for the hole concentration in the valence band of non-degenerate intrinsic semiconductor.

Now, for the Fermi level, as in case of intrinsic semiconductor,

number of electron concentration = number of hole concentration

$$2 \sqrt{\left(\frac{m_e K_B T}{2\pi\hbar^2}\right)^3} e^{-\frac{E_f - E_i}{K_B T}} = 2 \sqrt{\left(\frac{m_h K_B T}{2\pi\hbar^2}\right)^3} e^{-\frac{E_f - E_v}{K_B T}}$$

$$m_e^{\frac{3}{2}} e^{-\frac{E_c - E_f}{K_B T}} = m_h^{\frac{3}{2}} e^{-\frac{E_f - E_v}{K_B T}}$$

$$e^{\frac{E_f - E_v - E_c + E_f}{K_B T}} = \left(\frac{m_h}{m_e}\right)^{\frac{3}{2}}$$

$$e^{-\frac{E_c + E_v}{K_B T}} e^{\frac{2E_f}{K_B T}} = \left(\frac{m_h}{m_e}\right)^{\frac{3}{2}}$$

$$e^{\frac{2E_f}{K_B T}} = \left(\frac{m_h}{m_e}\right)^{\frac{3}{2}} e^{\frac{E_c + E_v}{K_B T}}$$

taking log on both sides;

$$\frac{2E_f}{K_B T} = \ln\left(\frac{m_h}{m_e}\right)^{\frac{3}{2}} + \frac{E_c + E_v}{K_B T}$$

$$E_f = \frac{3}{2} \ln\left(\frac{m_h}{m_e}\right) \frac{K_B T}{2} + \frac{(E_c + E_v)}{2}$$

$$E_f = \frac{(E_c + E_v)}{2} + \frac{3}{4} \ln\left(\frac{m_h}{m_e}\right) K_B T$$

by S.G. Sir

where, $E_f = E_i$ is Fermi energy for intrinsic semiconductor

This shows Fermi-energy level at room temperature is nearly above at the mid-way of the forbidden gap as mass of hole is slightly greater than mass of electron and $\frac{(E_c + E_v)}{2} \gg K_B T$ at room temperature.

where at absolute temperature $T=0K$, E_i fermi energy of intrinsic semiconductor exactly at mid-way between conduction and valence band.

$$\text{i.e. } E_i = \frac{(E_c + E_v)}{2}$$

In case of intrinsic Semiconductor both free electrons in conduction band and holes in valence band are use for charge conduction hence intrinsic charge density is given by

$$n_i^2 = np$$

$$n_i^2 = N_c N_v e^{-\frac{E_c - E_f}{K_B T}} e^{-\frac{E_f - E_v}{K_B T}}$$

$$n_i^2 = N_c N_v e^{-\frac{E_c - E_v}{K_B T}}$$

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_c - E_v}{2K_B T}}$$

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2K_B T}} \text{ where } (E_c - E_v) = E_g \text{ is band gap energy.}$$

This clearly shows that in intrinsic semiconductor, intrinsic carriers concentration depends only on band energy gap and absolute temperature but independent of Fermi energy level.

2. Calculation of carrier concentration of Extrinsic semiconductor:-

An Extrinsic semiconductors are of two types N-type and P-type.

a. Calculation of carrier concentration for N-type Extrinsic semiconductor:-

An N-type extrinsic semiconductor consists of pentavalent atoms or electron donor elements as doped impurities like phosphorous, Antimony, Arsenic etc. and hence has an electron as major charge carrier for electricity conduction. As the temperature is slightly increased above absolute temperature 0K, donor atoms gets ionized negatively and contribute additional electron to conduction band. Hence, in N-type extrinsic semiconductor, the electrons jump from valence band and that donated by donor atoms, all together take part in conduction and has higher conductivity than intrinsic semiconductor.

For the determination of carrier concentration at T temperature, let N_d be concentration of donors atom at E_d energy levels. Further, assuming that Fermi energy lying more closer to conduction band. Then, the electron concentration in conduction band is given by

$$n = N_c e^{-\frac{E_c - E_f}{K_B T}} \dots\dots\dots 1$$

Further, the electron concentration is express as sum of number of ionized donor atoms concentration (N_d^+) and hole concentration in valence band (p).

$$n = N_d^+ + p \dots\dots\dots 2$$

For, higher temperature, the number of electron concentration in conduction band is greater than that from valance band. So, p may neglected.

$$n = N_d^+ \dots\dots\dots 3$$

further for, concentration of ionized donor atom, it is assumed that N_d^+ number of donor atoms ionized out of N_d donor atoms at T temperature. Then,

$$\frac{N_d^+}{N_d} = f(E_d^+) \quad \text{by S.G. Sir}$$

where, $f(E_d^+)$ is Fermi Dirac statistics that gives probability of ionization of N_d^+ donor atoms at E_d energy level out of N_d .

$$\frac{N_d^+}{N_d} = 1 - f(E_d)$$

where $f(E_d)$ is Fermi Dirac statistics for non ionized donor atoms at E_d energy state.

$$N_d^+ = N_d \left[1 - \frac{1}{e^{-\frac{(E_f - E_d)}{K_B T}} + 1} \right]$$

$$N_d^+ = N_d \frac{e^{-\frac{(E_f - E_d)}{K_B T}}}{e^{-\frac{(E_f - E_d)}{K_B T}} + 1}$$

Since, $(E_f - E_d) \ll K_B T$, So

$$N_d^+ = N_d e^{-\frac{(E_f - E_d)}{K_B T}} \dots\dots\dots 4$$

Now, from equation 1, 4 and 3;

$$N_c e^{-\frac{(E_c - E_f)}{K_B T}} = N_d e^{-\frac{(E_f - E_d)}{K_B T}}$$

$$\frac{N_c}{N_d} = e^{-\frac{(E_c - E_d)}{K_B T}}$$

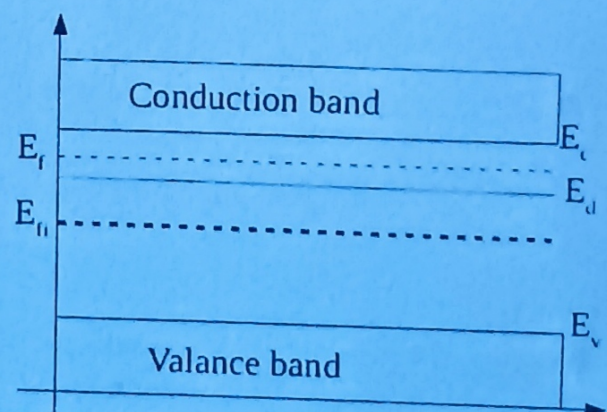


Fig: Fermi energy level (E_f) in N-type extrinsic SemiConductor that lies between E_c and E_d .

$$\frac{N_c}{N_d} = e^{\frac{(-2 E_f + E_c + E_d)}{K_B T}}$$

$$\ln\left(\frac{N_c}{N_d}\right) = \frac{-2 E_f + (E_c + E_d)}{K_B T}$$

$$K_B T \ln\left(\frac{N_c}{N_d}\right) - (E_c + E_d) = -2 E_f$$

$$E_f = \frac{K_B T}{-2} \ln\left(\frac{N_c}{N_d}\right) + \frac{(E_c + E_d)}{2}$$

$$E_f = \frac{K_B T}{2} \ln\left(\frac{N_d}{N_c}\right) + \frac{(E_c + E_d)}{2} \dots\dots\dots 5$$

This gives the position of Fermi level at moderate (room) temperature and shows position of Fermi energy level varies with temperature and concentration of donor atom doped in in N-type extrinsic semiconductor.

At low temperature i.e. at absolute temperature $T=0K$,

$$E_f = \frac{(E_c + E_d)}{2} \quad \text{by S.G. Sir}$$

Fermi level of N-type semiconductor lies in between donor atoms energy level and minimum energy level of conduction band.

Further, the electron carrier concentration in conduction band from equation 1 and 5, we get;

$$n = N_c e^{-\frac{E_c - \frac{K_B T}{2} \ln\left(\frac{N_d}{N_c}\right) - \frac{(E_c + E_d)}{2}}{K_B T}}$$

$$n = N_c e^{-\frac{\frac{(E_c - E_d)}{2} - \frac{K_B T}{2} \ln\left(\frac{N_d}{N_c}\right)}{K_B T}}$$

$$n = N_c e^{-\frac{(E_c - E_d)}{2 K_B T}} + e^{\frac{K_B T \ln\left(\frac{N_d}{N_c}\right)}{2 K_B T}}$$

$$n = N_c e^{-\frac{(E_c - E_d)}{2 K_B T}} + e^{\ln\left(\frac{N_d}{N_c}\right)^{\frac{1}{2}}}$$

$$n = N_c e^{-\frac{(E_c - E_d)}{2 K_B T}} + \left(\frac{N_d}{N_c}\right)^{\frac{1}{2}}$$

$$n = \sqrt{N_c N_d} e^{-\frac{(E_c - E_d)}{2 K_B T}}$$

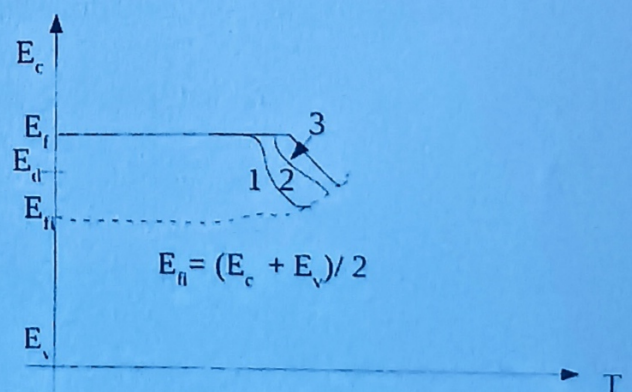


Fig: Dependence of Fermi level with temperature in N-type Semiconductor.

$$n = \sqrt{N_c N_d} e^{-\frac{\Delta E}{2k_B T}} \dots\dots\dots 7$$

where, $\Delta E = E_c - E_d$ is energy gap at $T=0K$

N_c = Concentration of electrons in conduction band

N_d = Concentration of donor atoms doped in semiconductor.

This is required expression for concentration of major charge carriers i.e electron on conduction band in N-type of extrinsic semiconductor at given room temperature $T K$.

This shows, at $T=0K$ N-type extrinsic semiconductor acts as insulator as major electron charge carrier concentration on conduction band drop to zero.

b. Calculation of carrier concentration for p-type Extrinsic semiconductor:-

A p-type extrinsic semiconductor consists of trivalent atoms or hole donor (electron acceptor) elements as doped impurities like Aluminium, Gallium, indium etc. and hence has a hole as major charge carrier for electricity. Holes are electron deficiency region which acts as +ve charge carriers. As the temperature is slightly increased above absolute temperature $0K$, these acceptor atoms gets ionized positively by acquiring electron in valence band and contribute additional p-type or hole conductance in valence band. Hence, in p-type extrinsic semiconductor, the hole presents in valence band initially before doping and holes created due to electrons jump from valence band to acceptor atom, all together involves in conduction and has higher conductivity than intrinsic semiconductor.

For the determination of hole carrier concentration at T temperature in p-type extrinsic semiconductor, let N_a be concentration of acceptor atoms at E_a energy levels. Further, assuming that Fermi energy lying more closer to valence band.

Then, the hole concentration in conduction band is given by

$$p = N_v e^{-\frac{E_f - E_v}{k_B T}} \dots\dots\dots 1$$

by S. G. Sir

Further, the hole concentration is express as sum of number of ionized acceptor atoms concentration (N_a^-) and electron concentration in conduction band (n).

$$p = N_a^- + n \dots\dots\dots 2$$

For, higher temperature, the number of hole concentration in valence band is greater due to acceptor atoms than that from electron concentration jump to conduction band. So, n may neglected.

$$p = N_a^- \dots\dots\dots 3$$

further for, concentration of ionized acceptor atom it is assumed that N_a^- number of acceptor atoms ionized out of N_a acceptor atoms at T temperature. Then,

$$\frac{N_a^-}{N_a} = f(E_a^-)$$

where, $f(E_a^-)$ is Fermi Dirac statistics that gives probability of ionization of N_a^- donor atoms at E_a energy level out of N_a .

$$N_a^- = N_a \left[1 - \frac{1}{e^{-\frac{(E_a - E_f)}{k_B T}} + 1} \right]$$

Since, $(E_a - E_f) \gg k_B T$, So

$$N_a^- = N_a e^{-\frac{(E_a - E_f)}{k_B T}} \dots\dots\dots 4$$

Now, from equation 1, 4 and 3;

$$N_v e^{-\frac{(E_f - E_v)}{k_B T}} = N_a e^{-\frac{(E_a - E_f)}{k_B T}}$$

$$\frac{N_v}{N_a} = e^{\frac{-(E_a - E_f)}{K_B T}} e^{\frac{(E_f - E_v)}{K_B T}}$$

$$\frac{N_v}{N_a} = e^{\frac{(2E_f - (E_a + E_v))}{K_B T}}$$

$$\ln\left(\frac{N_v}{N_a}\right) = \frac{2E_f - (E_a + E_v)}{K_B T}$$

$$K_B T \ln\left(\frac{N_v}{N_a}\right) + (E_a + E_v) = 2E_f$$

$$E_f = \frac{K_B T}{2} \ln\left(\frac{N_v}{N_a}\right) + \frac{(E_a + E_v)}{2}$$

$$E_f = \frac{(E_a + E_v)}{2} - \frac{K_B T}{2} \ln\left(\frac{N_a}{N_v}\right) \dots\dots\dots 5$$

This gives the position of Fermi level at moderate (room) temperature and shows position of Fermi energy level varies with temperature and concentration acceptor atoms doped in p-type extrinsic semiconductor.

At low temperature i.e. at absolute temperature $T=0K$,

$$E_f = \frac{(E_a + E_v)}{2}$$

by S.G Sir

Fermi level of N-type semiconductor lies in between acceptor atoms energy level and maximum energy level valence band.

Further, the hole carrier concentration in conduction band from equation 1 and 5, we get;

$$p = N_v e^{-\frac{\frac{(E_a + E_v)}{2} - \frac{K_B T}{2} \ln\left(\frac{N_a}{N_v}\right) - E_v}{K_B T}}$$

$$p = N_v e^{\frac{\frac{-(E_a - E_v)}{2} + \frac{K_B T}{2} \ln\left(\frac{N_a}{N_v}\right)}{K_B T}}$$

$$p = N_v e^{-\frac{(E_a - E_v)}{2K_B T} + e^{\frac{K_B T \ln\left(\frac{N_a}{N_v}\right)}{2K_B T}}$$

$$p = N_v e^{-\frac{(E_a - E_v)}{2K_B T} + e^{\ln\left(\frac{N_a}{N_v}\right)^{\frac{1}{2}}}}$$

$$p = N_v e^{-\frac{(E_a - E_v)}{2K_B T} + \left(\frac{N_a}{N_v}\right)^{\frac{1}{2}}}$$

$$p = \sqrt{N_v N_a} e^{-\frac{(E_a - E_v)}{2K_B T}}$$

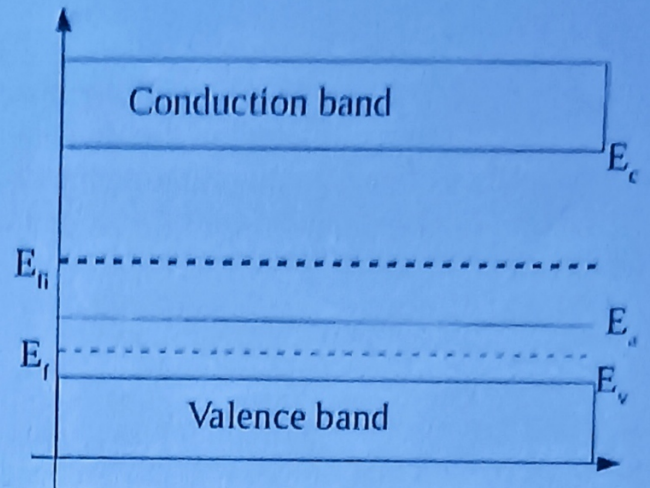


Fig: Fermi energy level (E_f) in p-type extrinsic semiconductor that lies between E_a and E_v .

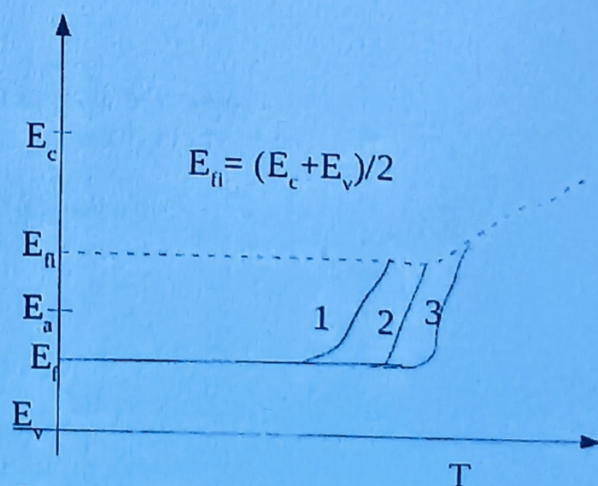


Fig: Dependence of Fermi level with temperature of P-type Semiconductor.

$$p = \sqrt{N_v N_a} e^{-\frac{\Delta E}{2K_B T}} \dots\dots\dots 7$$

where, $\Delta E = E_g - E_v$ is energy gap at $t=0K$

N_v = Concentration of hole in conduction band

N_a = Concentration of acceptor atoms doped in semiconductor.

This is required expression for concentration of major charge carriers

i.e hole on valence band in p-type of extrinsic semiconductor at given room temperature T K.

This shows at $T=0K$ p-type extrinsic semiconductor acts as insulator as major hole charge carrier concentration on valence band drop to zero.

Conductivity of an intrinsic Semiconductor :-

For intrinsic Semiconductor that is in pure state and has equal number of electron and holes as charge carriers. Further, for σ_e and σ_h be the electrical conductivity of electron and hole in intrinsic semiconductor respectively then the current density due to electrons and holes are given as;

$$J_e = \sigma_e E$$

$$J_h = \sigma_h E \dots\dots\dots 1$$

The ratio of drift velocity gain by charge carriers to that of applied electric field in conductor is define as mobility of charge carrier i.e.

$$\mu_e = v_{de} / E \text{ is mobility for electron [} v_{de} \text{ is drift velocity of electron]}$$

$$\mu_h = v_{dh} / E \text{ is mobility for electron [} v_{dh} \text{ is drift velocity of hole] } \dots\dots\dots 2$$

Further, the current density in terms of carrier concentration and its drift velocity is,

$$J_e = n e v_{de}$$

$$\text{or, } J_e = n e \mu_e E \quad [\text{ from eqn 2 }]$$

Similarly, for hole

$$J_h = p e \mu_h E \dots\dots\dots 3$$

Now, from eqn 1 and 3 ;

$$\sigma_e E = n e \mu_e E$$

$$\text{and } \sigma_h E = p e \mu_h E$$

$$\text{or, } \sigma_e = n e \mu_e$$

$$\text{and Similarly we get; } \sigma_h = p e \mu_h$$

The total conductivity in case of intrinsic semiconductor is due to both hole and electron hence;

$$\sigma = \sigma_e + \sigma_h$$

$$= n e \mu_e + p e \mu_h$$

$$\sigma = n e (\mu_e + \mu_h) \dots\dots\dots 5 \quad [\text{ since } n=p \text{ for intrinsic semi conductor}]$$

Further, the carrier concentration for intrinsic semiconductor is given by,

$$n_i = \sqrt{N_c N_v} e^{-\frac{E_g}{2K_B T}} \text{ where } (E_c - E_v) = E_g \text{ is band gap energy.}$$

$$N_v = 2 \sqrt{\left(\frac{m_h K_B T}{2\pi\hbar^2}\right)^3} \quad \& \quad N_c = 2 \sqrt{\left(\frac{m_e K_B T}{2\pi\hbar^2}\right)^3}$$

So, with this equation 5 becomes;

$$\sigma = e(\mu_e + \mu_h) \sqrt{N_c N_v} e^{-\frac{E_g}{2K_B T}}$$

$$\sigma = \sigma_0 e^{-\frac{E_g}{2K_B T}} \text{ where } \sigma_0 = e(\mu_e + \mu_h) \sqrt{N_c N_v}$$

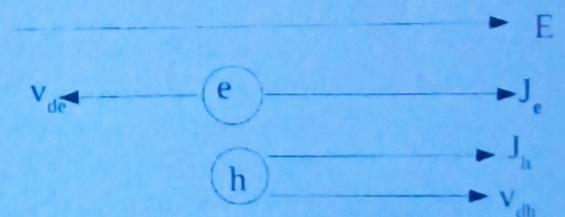


Fig: direction of E, current density and drift velocity for electron and hole inside semiconductor.

This shows, the conductivity of intrinsic semiconductor increases exponentially with temperature. As shown in figure,



Fig :variation of conductivity of intrinsic semi-conductor with temperature.

Numerical related to theses topics:

- The energy gap in silicon is 1.1 eV. The average effective mass is $0.31 m_e$. Calculate the electron concentration in the conduction band of silicon at room temperature 300K. Ans $2.6 \times 10^{15} \text{ m}^{-3}$.
- A sample of Si is doped with Phosphorous. The donor impurity level lies 0.045 eV below the bottom of the conduction band. At $T=300\text{K}$, E_f is 0.010 eV above the donor level. Calculate
 - the impurity concentration
Ans $2.79 \times 10^{24} \text{ m}^{-3}$
 - the number of ionized impurities
Ans $1.08 \times 10^{24} \text{ m}^{-3}$
 - the free electron concentration
Ans $1.08 \times 10^{24} \text{ m}^{-3}$
 - hole concentration [for Si, $E_g = 1.100 \text{ eV}$, $m_e^* = 0.31 m_e$ and $m_h^* = 0.38 m_h$] Ans $1.88 \times 10^6 \text{ m}^{-3}$
- The band gap in pure Germanium is $E_g = 0.67 \text{ eV}$.
 - Calculate the number of electrons per unit volume in the conduction band at 250K, 300K and at 350 K.
Ans $1.518 \times 10^{17} \text{ m}^{-3}$, $2.63 \times 10^{18} \text{ m}^{-3}$, $2.097 \times 10^{19} \text{ m}^{-3}$
 - Do the same for Silicon assuming $E_g = 1.1 \text{ eV}$. The effective mass of the electrons in germanium is $0.12 m$ and in Silicon $0.31 m$, where m is the free electron mass.
Ans $2.73 \times 10^{13} \text{ m}^{-3}$, $2.53 \times 10^{15} \text{ m}^{-3}$, $6.66 \times 10^{16} \text{ m}^{-3}$
- Suppose that the effective mass of hole in a material is four times that of electrons. At what temperature would the Fermi level shifted by 10% from the middle of the forbidden energy gap? Let, $E_g = 1 \text{ eV}$.
by S.G. Sir
Ans 558.1 K
- The energy gap in germanium is 0.67 eV. The electron and the hole effective masses are $0.12 m$ and $0.23 m$ respectively, where m is the mass of free electron. Calculate
 - the Fermi energy.
 - the electron density and
 - the hole density at $T=300\text{K}$
Ans 0.347 eV, $3.96 \times 10^{18} \text{ m}^{-3}$, $1.05 \times 10^{19} \text{ m}^{-3}$
- A certain intrinsic Semiconductor has a band gap E_g is 0.2 eV. Measurement shows that it has a resistivity at room temperature 300K of $0.3 \Omega\text{m}$. What would you predict its resistivity to be at 350K?
Ans $0.136 \Omega\text{m}$
- The resistivity of intrinsic Germanium at 300K is $47 \Omega\text{cm}$. Calculate the intrinsic carrier density of germanium at the same temperature, given that the electron and hole mobilities are $0.39 \text{ m}^2\text{V}^{-1}\text{S}^{-1}$ and $0.19 \text{ m}^2\text{V}^{-1}\text{S}^{-1}$ respectively.
Ans $2.29 \times 10^{19} \text{ m}^{-3}$